INCH-POUND

MIL-S-48685 (AR)
AMENDMENT 2
16 March 1994
SUPERSEDING
AMENDMENT 1
16 February 1990

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143E1
ELECTRICAL PARTS AND ASSEMBLIES

This amendment forms a part of Military Specification MIL-S-48685, dated 4 February 1987, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Defense.

PAGE 2

* 2.1.1, under STANDARDS, delete: "MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For" and substitute "MIL-STD-2000 - Soldered Electrical and Electronic Assemblies, Standard Requirements For"

PAGE 3

3.3.3.1, line 5: Delete "2.5" and substitute "1.5".

PAGE 4

* 3.3.4.2, line 2: Delete "5.0 ± 0.25V" and substitute "4.50 ± 0.25 V".

PAGE 5

3.4.1, Table II, Terminal E20 to E21, under requirements:

Delete "Less than 19 MEG: and substitute "Less than 500K".

3.4.1, Table II, Terminal E21 to E20, under requirements:

Delete "Greater than 20 MEG: and substitute "Less than 500K".

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PAGE 9

J1-13, line 2, Under points requiring logic high: Delete "J17" and substitute "J1-17".

PAGE 10

J1-27, Under points requiring logic high: Delete "FR12" and substitute "FL2".

J1-33, Under points requiring logic high: Add "L20".

CASE Ground
(Hole #182), Under points requiring logic high: Delete "All low" and substitute "J1-50, L5".

PAGE 15

Table IX, g&h: Delete "2.5 (six places)" and substitute "4.0 (six places)".

PAGE 16

3.7.1, line 7: Delete "2.5" and substitute "1.5".

3.7.2, line 6: Delete "2.5" and substitute "4.0".

* 3.7.4: Delete in its entirety and substitute

"3.7.4 Fire circuit function. The initiation of either the left or right fire circuit functioning is determined by the occurrence of the fire TM output changing from 0 ± .5V to 4.5 ± 1.0V at (+) J1-20 to J1-15. The left fire circuit has completed functioning when the left fire circuit capacitor voltage (+) J1-38 to (-) J1-15 has discharged to less than 15.0 VDC in no greater than 400 milliseconds after the the occurrence of a fire TM output. The right fire circuit has completed functioning when the right fire circuit circuit capacitor voltage (+) J1-36 to (-) J1-15 has discharged to less than 15.0 VDC in no greater than 400 msec after the occurrence of a fire TM output.

Either one or two Fire TM pulses can be issued. A single Fire TM pulse is issued when both Fire Circuits function simultaneously or within a short time of one another. Two separate Fire TM pulses indicate that simultaneous functioning has not occurred. In this case the initial Fire TM pulse shall be the reference point for the measurement of the time required for the discharge of the fire circuit (left or right) whose functioning caused the initial Fire TM pulse. The occurrence of the second Fire TM pulse shall be the reference point for the measurement of the time required for the remaining non-functioned Fire Circuit (left or right) to discharge."

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PAGE 17

3.7.4.1.1: Delete in its entirety and substitute

"3.7.4.1.1 Loss of power after arming. The initiation of fire circuit functioning as defined by paragraph 3.7.4 shall occur in no less than 10 milliseconds and no greater than 50 milliseconds upon removal of power/destruct (+) J1-30 to (-) J1-15."

3.7.4.1.2: Delete in its entirety and substitute

"3.7.4.1.2 Fire upon arming after loss of power. The fire circuits shall function as described in paragraph 3.7.4 upon arming, after removal of power/ destruct (+) J1-30 to (-) J1-15."

3.7.4.2.1, Line 2: Delete "5.0 ± 0.25V" and substitute "4.5 ± 0.25V".

PAGE 20

3.9.4: Delete in its entirety and substitute

"3.9.4 Semiconductor Burn-In."

Add:

"3.9.4.1 Diodes and Rectifiers. Devices shall be subjected to burn-in testing in accordance with the details herein and MIL-STD-750, Method 1038, Test Condition B and shall meet the final electrical requirements after exposure.

3.9.4.2 Transistors. Devices shall be subjected to burn-in testing in accordance with details herein and MIL-STD-750, Method 1039, Test condition B and shall meet the final electrical requirements after exposure."

PAGE 22

4.3.1, Under drawings: Delete "12630705, 12630850 and 12630854" in its entirety.

4.3.1, Under drawings: Delete "12630851," and substitute "9311361".

4.3.3, Line 1: Delete "componment" and substitute "component".

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* 4.4.2.4, Sheet 1 of 1, Major 101, under Examination or test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 31

* 4.4.2.7, Sheet 1 of 1, Major 101, under Examination or test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 33

* 4.4.2.8, Sheet 1 of 1, Major 101, under Examination or test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 35

* 4.4.2.9, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 36

* 4.4.2.9.1, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 37

* 4.4.2.10, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 38

4.4.2.10.1, Title: Add "Destructive Testing".

4.4.2.10.1, Drawing Number: Add "-1".

4.4.2.10.1, Notes: Add "1/ not required for 12630838-2".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Bridge Rectifier				12630860
					NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED (TSR)				
Critical	None defined				
Major					
101	Temperature cycling		100%	3.9.1	Test
102	Burn-In		100%	3.9.4	Test
103 1/	Forward voltage drop per leg		100%	3.2	Test
104 1/	Leakage current per leg		100%	3.2	Test
Minor					
201	Marking (after final electrical)		0.65%	3.10	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

1/ The percent defective allowable (PDA) for the ELECTRICAL INSPECTIONS following temperature cycling and Burn-In shall not exceed 15 % of all units tested.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9.1	Bridge Rectifier (Environmental, Destructive Test)				12630860
					NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TEST SEQUENCE REQUIRED (TSR)				
Major					
101	Varible frequency vibration (non-operating)		0.40%		MIL-STD-750 (METHOD 2056)
102	Constant acceleration		0.40	3.9.2	Test
103	Mechanical shock		0.40	3.9.6	Test
104	Forward voltage drop per leg		0.40	3.2	Test
105	Leakage current per leg		0.40	3.2	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Transistor, High Power, Silicon P.N.P 2N6193 (-1) or JANTX2N6193 (-2)	1 OF 1	12630838/-1/-2/1/
			NEXT HIGHER ASSEMBLY 12630829

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.9.4	Test
102	Collector cut-off current with emitter open		100%	3.2	Test
103	Emitter-cut-off current		100%	3.2	Test
104	Collector-to-emitter sustaining voltage with base open		100%	3.2	Test
105	Collector-to-emitter saturation voltage		100%	3.2	Test
106	Base-to-emitter voltage		100%	3.2	Test
107 1/	DC forward current transfer ratio		100%	3.2	Test
Minor					
201 1/	Marking (after final electrical)		0.65%	3.10	Visual
202 1/	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

1/ For JANTX2N6193 MIL-STD-19500 1561 applies and only category 107, 201, & 202 are required.

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4.4.2.11, Sheet 2 of 2, next higher assembly: Delete "12630854" and substitute "12630849".

PAGE 40

4.4.2.12, Sheet 1 of 1, drawing number: Delete "12630851" and substitute "9311361".

PAGE 44

* 4.4.2.15, Sheet 1 of 1, Major 101 under Examination or Test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 45

4.4.2.15, Sheet 1 of 2, paragraph: Delete "4.4.2.15" and substitute "4.4.2.15.1".

PAGE 46

4.4.2.15, Sheet 1 of 1, paragraph: Delete "4.4.2.15" and substitute "4.4.2.15.1".

PAGE 48

* 4.4.2.17, Sheet 1 of 1, Major 101 under Examination or Test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

PAGE 50

4.4.2.21, Sheet 1 of 1, next higher assembly: Delete "12630854" and substitute "12630849".

PAGE 51

4.4.2.22, Sheet 1 of 1: Delete in its entirety.

PAGE 52

* 4.4.2.23, Sheet 1 of 1, Major 101 under Examination or Test, Delete: "MIL-STD-1460" and substitute "MIL-STD-2000".

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4.4.2.26, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	Electrical Assembly Electrical Operation				12630828
					NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fire circuit charging (Power/Destruct)		100%	3.7.2	Test
102	Fire upon arming after loss of power		100%	3.7.4.1/ 3.7.4	4.5.2/Test

NOTES:

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4.4.2.27, Sheet 1 of 1, major 113, under requirement paragraph: Add "3.7.4".

PAGE 57

4.5.1.1,a.: Delete "Armed and Locked" and substitute "Safe and Locked".

4.5.1.2 Delete in its entirety and substitute:

"4.5.1.2 Dynamic conditions.

a. Electrical assembly Armed and Locked.

b. Remove 31.0 +0.5VDC on Power/Destruct.

c. Measure time to occurrence of fire TM (3.7.4.1.1).

d. Measure time for L&R fire circuits to function (3.7.4)."

4.5.2.2,c.: Delete sentence and substitute "Measure time for the fire circuits to function (3.7.4.1.2)."

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